

An Open-Source Cost-Effective Arduino Platform for High-Throughput Solar Device Stability Testing

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Abstract

Perovskite solar devices have achieved high power conversion efficiencies, but their long-term stability remains a challenge to their widespread commercial adoption. Therefore, ubiquitous stability testing by researchers around the world is critical to understanding not just how efficient a device is but also how durable. However, the high cost of commercial stability analyzers presents a significant barrier for many research laboratories. We have developed an open-source stability measurement system (branded here as SMS) based on the Arduino platform. This system provides essential measurement capabilities at approximately 1% of the cost of commercial alternatives. Each \$100 SMS module enables simultaneous current-voltage (JV) characterization and maximum power point tracking (MPPT) on eight separate solar devices. The system’s modular design allows for expansion to monitor over 250 devices concurrently. The SMS consists of custom printed circuit boards, 3D-printed sample holders, and Python-based control software. Validation against a commercial system demonstrated a relative error of 0.76% for JV derived power conversion efficiency. A 200-hour stability test confirmed the system’s reliability for long-term studies. While variability in contact resistance limits the system’s precision for absolute benchmarking, the SMS is effective as a screening platform for comparative stability studies. By providing greater access to high-throughput aging studies, this open-source platform can facilitate broader exploration of perovskite compositions and accelerate research toward commercial viability.

1 Introduction

Perovskite solar devices (PSCs) have advanced rapidly, with power conversion efficiencies (PCEs) increasing from approximately 4% in 2009 to over 26% within a decade. They are now rivaling the performance of silicon-based devices while offering simpler and more cost-effective manufacturing processes. [1,2] However, their limited operational stability remains a significant obstacle to commercialization. Various factors such as illumination, moisture, oxygen, thermal stress, and electrical bias can induce ion migration, interfacial reactions, and phase segregation, all of which contribute to degradation of the solar device. [3–7]

To address these stability challenges, specialized instrumentation is required to apply stress conditions and perform electrical measurements. However, commercial stability analyzers are expensive, with costs ranging from \$10,000 to \$60,000. [8] Lower-cost commercial systems exist, but are often limited to single device measurements. [9] A notable high-throughput system developed by Köbler et al. offers parallel maximum power point tracking (MPPT) across hundreds of perovskite devices; however, its bespoke design and unpublished assembly details make it difficult for other research groups to reproduce. [10] These factors restrict the ability of many research groups to conduct comprehensive studies on combinatorial compositional spaces or to obtain statistically significant data from large substrate populations.

In recent years, the open-source hardware movement has provided a practical approach to making high-throughput stability testing more accessible. Community-driven initiatives have produced affordable environmental chambers, LED-based solar simulators, and optical monitoring systems. [8,11,12] Although some low-cost, open-source systems for single-

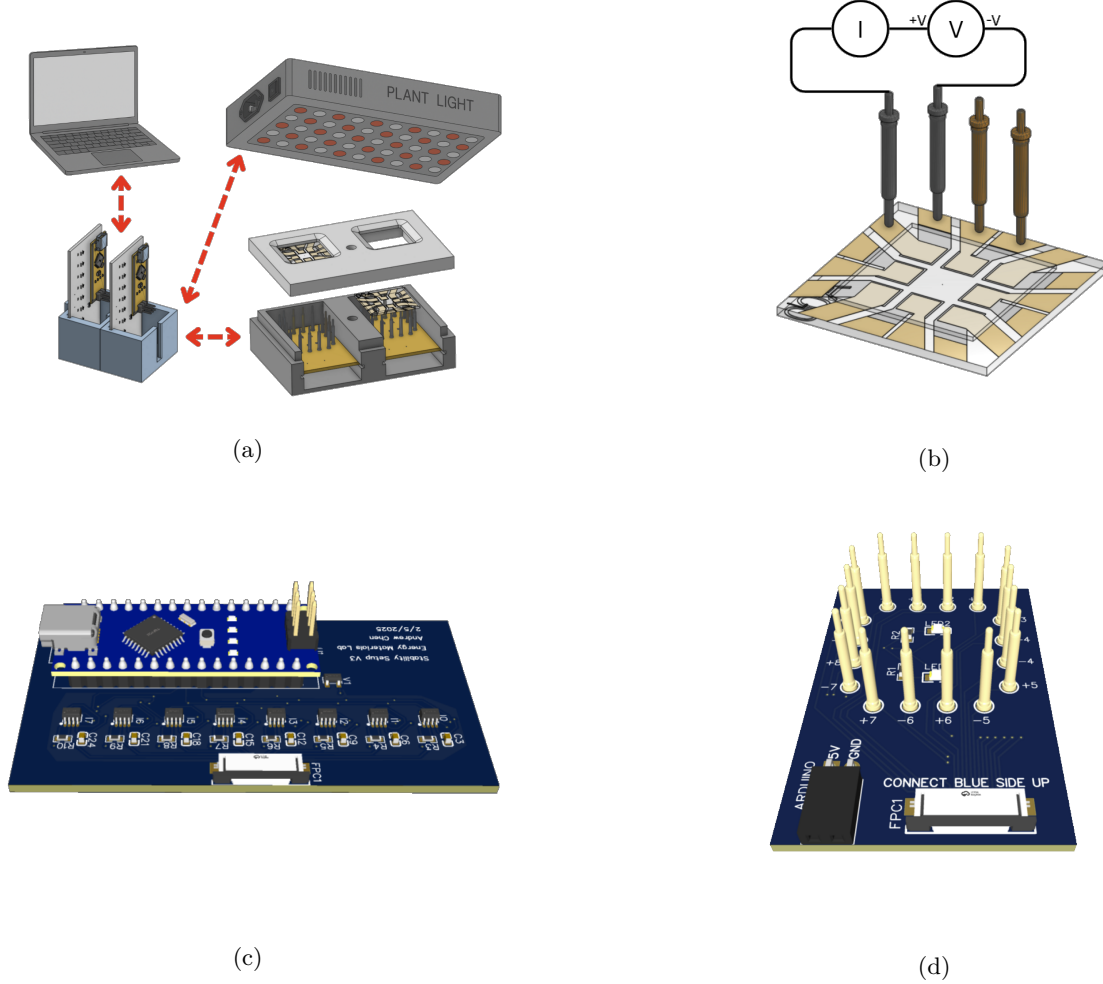


Figure 1: Schematic visualization of the Stability Measurement System (SMS). a) Overall system block diagram. Red arrows indicate data and power connections. b) Detail of the pogo-pin contacts in the 3D-printed enclosure. c) Main PCB assembly: eight independent, electrically isolated measurement channels. d) Sample holder mating with eight devices via spring-loaded pogo pins.

channel current-voltage (JV) and MPPT have been developed, they often require specialized knowledge to reproduce and operate. [13–15] Currently, no open-source instrument is capable of performing automated, parallel MPPT and JV measurements across multiple devices.

In this work, we introduce an easily reproducible \$100 (per measured 8 devices at time of writing) Arduino-based SMS designed to fill this gap. Each SMS module includes a custom printed circuit board (PCB), a sample holder, a 3D-printed enclosure, and Python-based control software with a graphical user interface (GUI) to perform parallel JV and MPPT measurements on eight independent pixels. Multiple modules can be connected to a single Windows computer, enabling the monitoring of over 250 de-

vices in parallel. The system requires only basic soldering, 3D printing, and computer skills for assembly. Some minor PCB design skills will be needed if the solar cell layout is different from the one used in this paper. The SMS provides the core functionality of commercial analyzers at approximately 1% of the cost. To benchmark the SMS against a commercial system, we have measured the same 32 devices across both systems, and compared the measured fill factor (FF), open-circuit voltage (V_{OC}), short-circuit current density (J_{SC}), and maximum power conversion efficiency derived from MPPT (PCE_{MPPT}). We have also demonstrated its reliable operation over a 200-hour period and released all hardware and software under an open-source license to encourage community adoption and further development [16].

2 System Description

At the core of each SMS module is a two-layer PCB (Fig. 1c). A custom PCB is essential for integrating the numerous components into a compact, reliable, and factory-reproducible measurement system, which would be difficult to achieve with standard prototyping boards. This approach minimizes potential points of failure and electrical noise, while ensuring the cost-effective reproduction of identical modules. The board includes an Arduino microcontroller for sequencing and USB communication, eight 12-bit Digital-to-Analog Converters (DACs) that supply a programmable bias voltage (0–3.3 V, 0.8 mV steps), and eight dual-channel Analog-to-Digital Converters (ADCs) for measuring voltage (−0.3–26 V, 4 mV resolution) and current (0–32 mA, 8 μ A resolution). Each of the eight channels is fully isolated, allowing for simultaneous JV sweeps or MPPT on all devices. Each channel operates independently with its own dedicated DAC output and ADC input, which prevents electrical crosstalk between measurements. This isolation ensures that the electrical characteristics of one device do not affect the measurements of adjacent channels, enabling true parallel testing.

The device holder is a companion PCB that connects to the main board via a 16-pin ribbon cable, with two conductors per channel. Two additional breadboard wires power built-in LEDs that can be toggled for mask alignment. Each device is contacted by spring-loaded pogo pins within a 3D-printed frame (Fig. 1a), which ensures consistent pressure on both electrodes. The sample holder was connected to the SMS using 16-pin, 0.5 mm pitch flat flexible cables (FFC) that snap into the integrated connector. With a 600 mm FFC, the resistance of each channel is 2 Ω between the pogo pin tip and the SMS input/output. The 3D-printed frame also includes a press-fit port for a 2.5 mm internal diameter N₂ line. Illumination is provided by a commercially available LED grow light. Power and data are transmitted over a single USB cable to a host PC. System capacity can be expanded by connecting additional SMS modules via USB. A powered USB hub is recommended when using more than four modules per computer USB port, as a standard USB port is not capable of supplying power for more than 4 SMS systems at a time. We have verified stable parallel data collection from eight SMS systems (64 devices) on a Windows 10/11 desktop, although the Python based software may run on Linux and MacOS based platforms with minimal modification. A minimum of 12 GB of RAM is recommended, and any modern CPU should be sufficient. The scalability of the SMS system is limited by the number of avail-

able USB host controllers, which varies significantly between computer models. A single USB controller has a theoretical capacity of 32 downstream devices (USB hubs are included in this count) [17]. Therefore, a modern computer can be expected to support a load of at least 20 to 30 SMS systems.

2.1 Software

The SMS is controlled by a Python-based GUI that runs on Windows 10/11 (Fig. 2). The software allows users to configure and run JV and MPPT measurements, view real-time data, and log results to CSV files. Key features include interactive configuration of scan parameters (e.g., voltage range, step size, sampling rate), a test queue for automated saving of test sequences, real-time plotting of JV curves and MPPT power traces, and continuous CSV logging for robust data archiving, which protects against data loss in the event of a PC crash. The GUI is designed to be intuitive and user-friendly, and a comprehensive user guide is available in the open-source repository [16].

JV Scan

To characterize initial performance and detect hysteresis, the system conducts a bidirectional voltage sweep, during which several key parameters can be configured. The scan range (0–3.3 V) defines the voltage span of the sweep, while the scan step size (0.001–1 V) sets the voltage increment between readings. The device area (cm²) is used to calculate current density from the measured current. To improve signal quality, the number of readings averaged per voltage step can be adjusted to reduce noise. Finally, the scan rate (mV/s) controls the speed of the voltage sweep, which is critical for accurately capturing the device’s JV characteristics.

MPPT Algorithm

We implemented a perturb-and-observe (P&O) tracking algorithm [18]. The initial voltage can be set manually or automatically from a prior JV scan, which sets the starting voltage at a percentage of V_{OC} [19]. Users can configure several parameters to control the tracking process: the voltage step size (0.001–1 V) determines the magnitude of the voltage perturbation at each step; the device area (cm²) is used for power conversion efficiency (PCE) calculations; the number of measurements per step allows for signal averaging to reduce noise; a measurement delay (ms) can be set to allow the device to stabilize after a voltage change; the measurement interval (ms) defines the time over which readings are taken for each voltage step; and the total measurement time can be specified in minutes or hours to define the experiment’s duration.

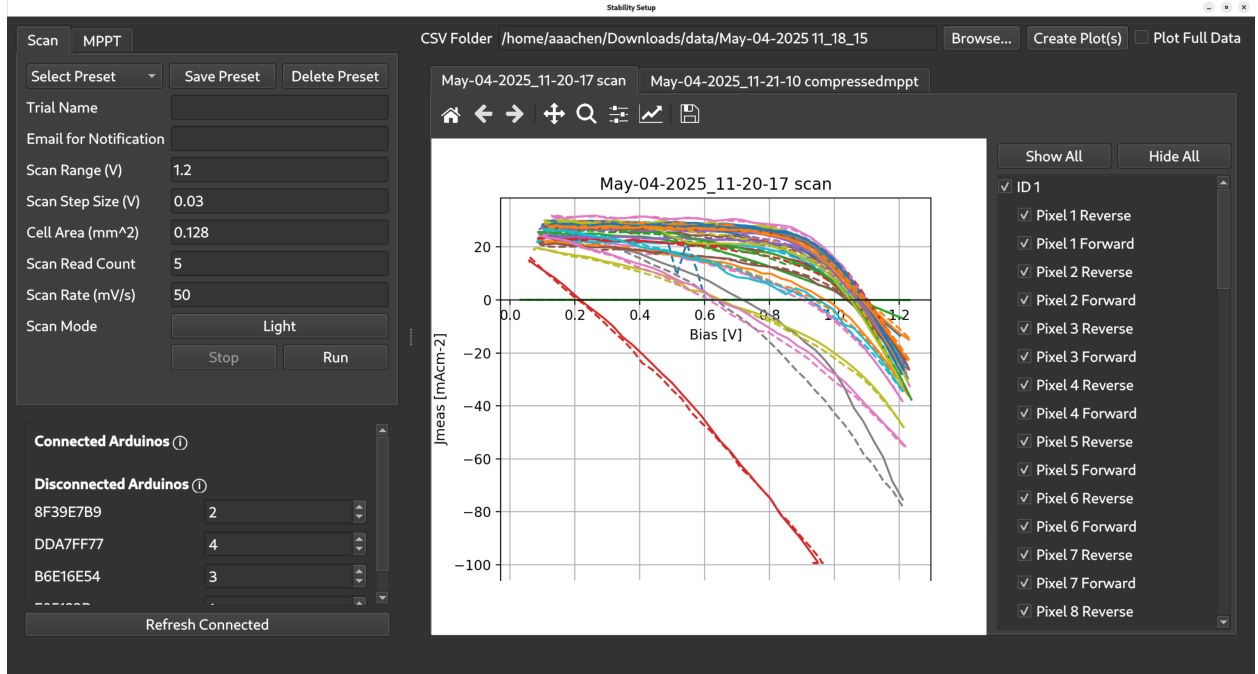


Figure 2: The graphical user interface (GUI) for the SMS, showing real-time results from a 5-minute MPPT measurement conducted on 32 perovskite solar devices simultaneously. The interface displays key metrics and allows for parallel data tracking. While the current software has a more feature rich, multi-page layout, this earlier version is presented to clearly show the system’s essential functions in a single view.

The inclusion of a user-configurable measurement delay, or settling time, is justified by measurement challenges specific to perovskite solar devices. Park et al. note that many PSCs are “slow-responding” and exhibit “transient hysteresis” in their I–V characteristics, a behavior commonly attributed to ionic migration [20]. Because ionic migration is slow, the photocurrent can require time to reach a steady value after a voltage change. Implementing a variable delay based on this principle allows our algorithm to adapt to different device types, helping it converge to the true steady-state maximum power point and avoid misleading transient readings.

The P&O method was selected for its straightforward implementation and low computational demand, which is well-suited for Arduino-based systems. In contrast to methods such as incremental conductance or fuzzy logic control, P&O does not require extensive calibration or complex algorithms, which improves its reliability. Since our experiments are conducted under stable laboratory conditions, the risk of common P&O issues, such as oscillation around the MPP or convergence to local minima, is minimal. This makes P&O a practical and reliable choice for our testing setup [18].

2.2 Cost and Assembly

The SMS system is engineered to make reproducing it straightforward. Complete assembly instructions and design files for both the PCBs and the 3D holder are provided in the open-source repository [16]. Links to potential providers are also provided at the same location. At present, the price of the setup is around \$8 per measured device [21], which is less than 1% of the cost of commercially available stability measurement platforms on the market. This price does not include the solar simulator. Ordering the PCBs through JLC PCB requires ordering 5 at a time, totaling around \$300 before taxes and shipping. The CAD files for the PCBs, device holder, and SMS case can be found in the supporting materials [22] [23] [24]. The code for the project, along with assembly instructions and links to all components, can be found in our open source repository [16].

3 Experimental

This section details the materials, methods, and protocols used to validate the custom-built solar measurement system (SMS) against a commercial system. The experiments were designed to assess the accu-

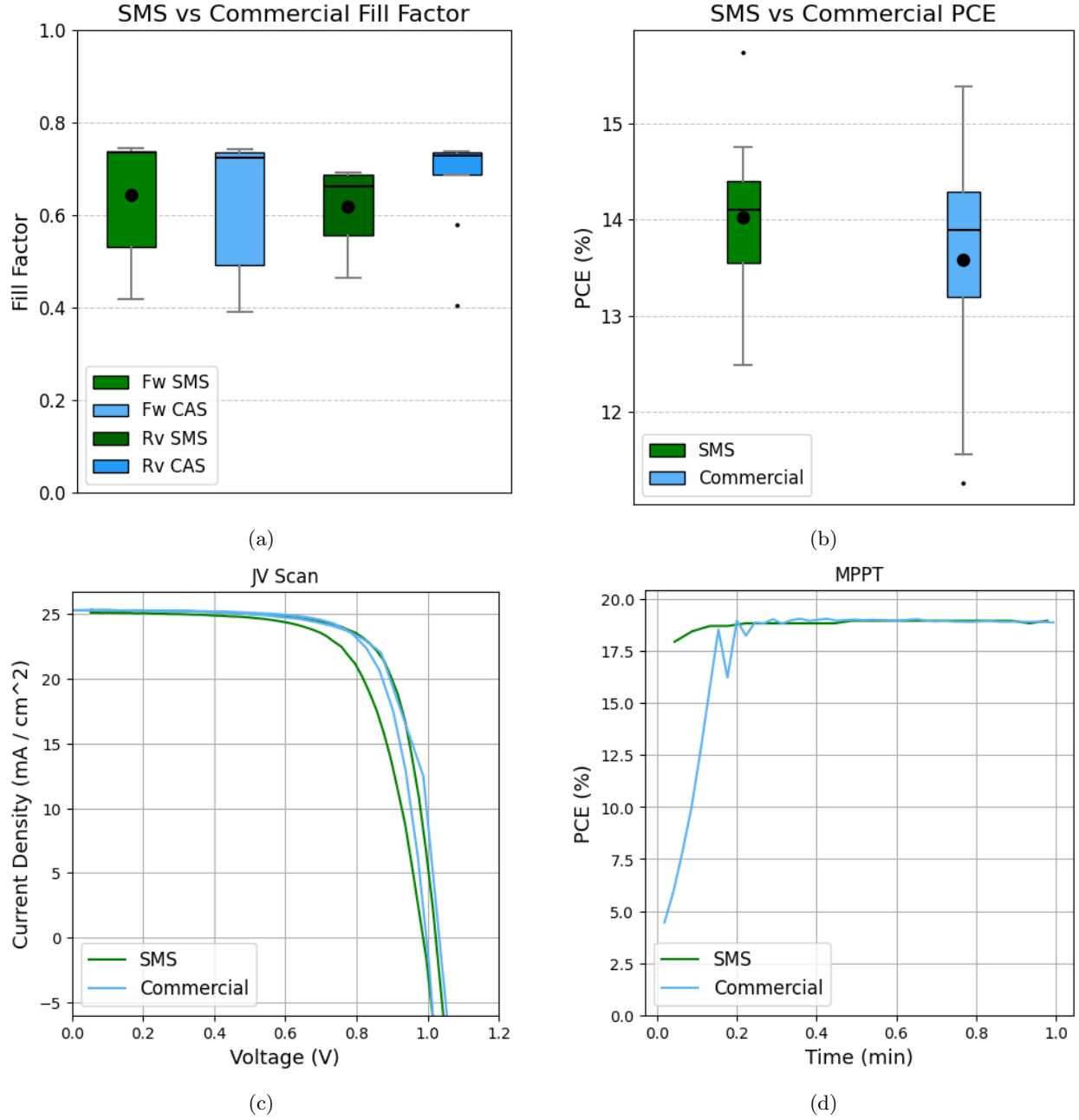


Figure 3: (a) Box plot of SMS vs commercial system Fill Factor on one substrate, with forward (Fw) and reverse (Rv) JV scans separated. (b) Box plot of SMS vs commercial system PCE on one substrate. (c) JV scan of one device from SMS plotted against commercial system with substrate in commercial system holder. (d) MPPT plot of one device from SMS plotted against commercial system with substrate in commercial system holder. In this plot, the SMS reaches stability before the commercial system because of the capability to start at a percentage of the V_{OC} .

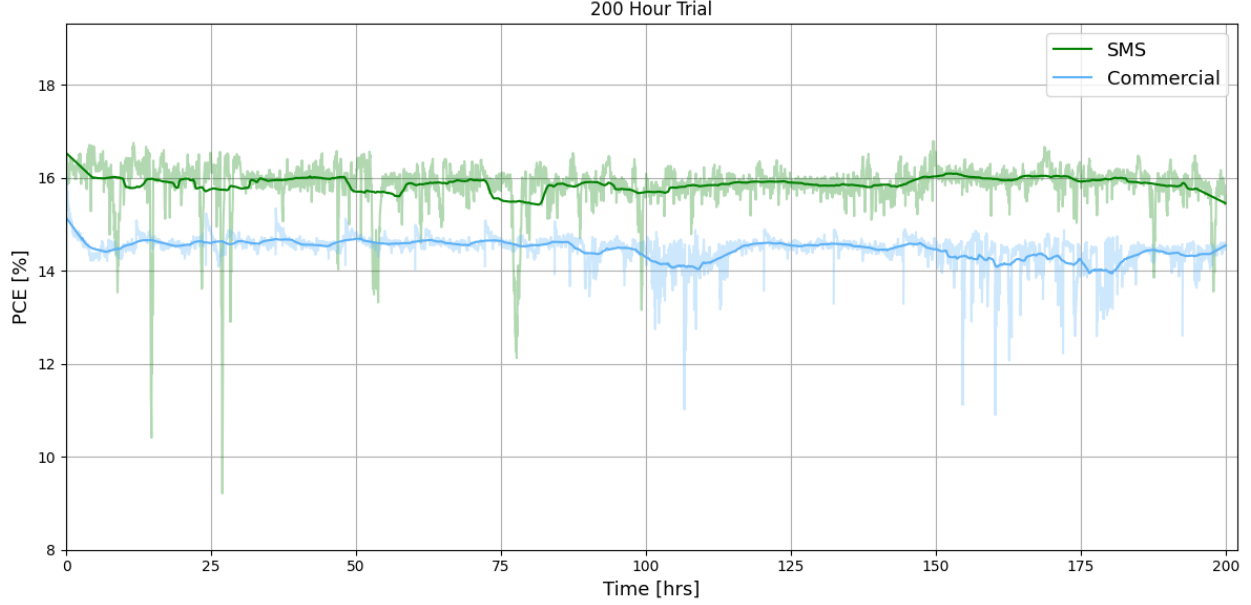


Figure 4: 200-hour stability measurement, plotting two separate devices from the SMS and a sample from the commercial system. This shows that the SMS can produce similar long term efficiency measurements to the commercial system.

racy, precision, and long-term stability measurement capabilities of the SMS. All JV and MPPT comparison measurements between the SMS and the commercial system were conducted using a WaveLabs Sinus 70 solar simulator [25], which was calibrated to 1 sun (100 mW/cm^2) with an AM 1.5G spectrum. For stability tests, a cost-effective LED grow light was used, and its intensity was adjusted to be 1 sun equivalent, as confirmed by a calibrated reference solar device. Four reference perovskite solar devices (PSCs), that had eight solar cells per substrate, were characterized (32 devices total). The substrates followed the n-i-p stack architecture reported by Kim et al. [26]: FTO / compact TiO_2 / mesoporous TiO_2 / PEAI interlayer / $\text{Cs}_{0.09}\text{FA}_{0.91}\text{PbI}_3$ perovskite / PEAI interlayer / Li-TFSI, tBP-doped spiro-OMeTAD / Au. Each device had an active area of approximately 0.128 cm^2 . Silver paste was applied to the device contacts to ensure consistent electrical connection during long-term testing. The primary metrics of interest were the FF, V_{OC} , J_{SC} , PCE_{MPPT} . The FF, J_{SC} , and V_{OC} were determined from the JV scans, while the PCE_{MPPT} was calculated from the stabilized output of the MPPT traces.

The following protocols were designed to establish the accuracy and precision of the SMS relative to a commercial standard. To validate the baseline accuracy of the SMS, a silicon diode was measured on both the SMS and the commercial system. The diode was

placed in each system's holder, and JV scans were conducted from 0 V to 1.2 V at a rate of 50 mV/s for comparison. To assess complete end to end system performance of the SMS, four PSC substrates with 8 devices each were measured sequentially on the commercial system and the SMS using their respective sample holders. The commercial holder was measured first. Substrates were then transferred from the SMS to the commercial system, with less than 10 minutes between measurements. Both systems were purged with nitrogen. Optical masks were not used to avoid potential misalignment when transferring samples that would change the active area. The substrates were briefly exposed to oxygen during handling. Both measurements included a forward JV sweep (0 V to 1.2 V), a reverse JV sweep (1.2 V to 0 V) at 50 mV/s, and a 1-minute MPPT trial. The MPPT trial was initiated at 85% of the V_{OC} from the JV scan. For the SMS, the voltage step sizes were 0.03 V for the JV scan and 0.005 V for the MPPT, while the commercial system used internally varied step sizes. PCE_{MPPT} was calculated from the average of the final 30 s of the MPPT trace.

To isolate the performance of the measurement electronics from variables related to sample handling and the SMS sample holder, an adapter board was used to connect the SMS directly to the commercial system sample holder. Four PSC substrates were tested using this integrated setup. In this experi-

ment, optical masks were used, reducing the active area to 0.0625 cm^2 . The JV and MPPT measurement parameters were identical to those in the Complete System Validation. To quantify the precision of the measurement system, four complete JV scans and MPPT measurements were recorded for four PSC substrates. The relative standard deviation (RSD) was calculated for the FF from the JV data and for the stabilized PCE_{MPPT} from the MPPT data to assess measurement consistency. The JV and MPPT measurement parameters were identical to those in the Complete System Validation. A 200-hour degradation test was performed on 16 parallel solar devices under continuous illumination from the LED grow light to assess reliability and long term performance of the SMS. Although the LED spectrum did not fully match the AM 1.5G solar spectrum, it provided controlled conditions for assessing degradation and stability. A Savitzky-Golay filter was applied to the resulting data to reduce noise and clarify degradation trends.

4 Results and Discussion

Silicon-Diode Baseline Validation Results

The baseline accuracy of the SMS was validated against the commercial system using a stable silicon diode, with the results summarized in Table 1. The measurements showed strong agreement in V_{OC} and JV-derived PCE, with relative errors of less than 1.5%. While larger deviations were observed for FF and J_{SC} , the overall results confirm that the SMS is sufficiently accurate for its intended purpose of long-term stability monitoring and rapid screening, where relative performance trends are more critical than absolute precision.

Complete System Validation Results

The JV characterization of 32 PSCs, summarized in Table 2, reveals important distinctions between systematic bias and random measurement error. The mean difference in J_{SC} ($+1.32 \text{ mA cm}^{-2}$) was statistically significant, indicating a systematic overestimation by the SMS compared to the commercial system. This bias may stem from differing internal resistances in the two systems, an offset in the current reading, and/or the exact height and probe spring pressure with which the devices were measured. In contrast, the mean differences for V_{OC} ($+24.4 \text{ mV}$) and FF (-0.55%) were not statistically significant, suggesting negligible systematic bias for these parameters.

However, the 95% limits of agreement, which define the expected range for 95% of measurement differences, were broad for all parameters. For instance,

the FF could vary by as much as $\pm 23.5\%$ between the two systems. This high degree of random variability, rather than systematic bias, appears to be the primary contributor to measurement discrepancy. Such scatter is likely attributable to inconsistent contact resistance from the pogo-pins, as the electrical characteristics of PSCs are known to be highly sensitive to applied pressure. This interpretation is supported by the work of Shi et al., who reported that variations in applied pressure alone can alter measured efficiency by as much as 7% [27]. The 3D-printed sample holder and pogo-pin assembly provides much less uniform contact pressure compared to the machined aluminum CAS holder. This was a design flaw that should be addressed in a further iteration, but were outside of the scope of this work. These findings underscore that while the SMS does not introduce substantial systematic error in key parameters like FF and V_{OC} , its precision is limited by random variability in the electrical contact.

MPPT trials showed a statistically insignificant mean PCE_{MPPT} offset of $+0.46\%$, but the limits of agreement ($\pm 5.16\%$) confirm the substantial random variability observed in the JV measurements. This level of precision is sufficient for long-term degradation studies, where changes in PCE_{MPPT} often exceed 10-20%, but may not be adequate for detailed mechanistic studies requiring higher accuracy.

The significant random variability observed between the SMS and commercial systems highlights the challenges that inherently come with measuring perovskites. A study by Dunbar et al. investigating inter-laboratory measurement reliability found that variability for perovskite devices can be nearly 10 times larger than for conventional silicon devices [28]. They demonstrated that for slowly responding perovskite devices, the specific choice of measurement method—including factors like scan rate and pre-conditioning—was a far greater source of discrepancy than equipment calibration or temperature control. While our study used consistent JV scan parameters between systems, it is entirely possible that the systems did not produce the exact same parameters. Differences in the system electronics or pre-measurement routines between the SMS and the commercial system could contribute to the observed scatter, compounding the contact variability issues.

Measurement Electronics Validation Results

To isolate the error contribution from the sample holder, measurements were repeated using the commercial system holder on both systems. As shown in Table 3, this modification significantly improved measurement agreement. The limits of agreement for FF narrowed from nearly $\pm 24\%$ to approximately

Table 1: Baseline accuracy of the SMS compared to the commercial system using a silicon diode.

Metric	Absolute Difference	Relative Error (%)
PCE (JV)	0.010	0.76
FF	0.035	6.21
J_{SC}	0.128 mA	5.74
V_{OC}	8 mV	1.38

Table 2: Agreement statistics for SMS vs. commercial system (Each system with its respective holder).

Metric	Mean Difference	MAE	RMSE	95% CI	95% Limits of Agreement
Fill factor (%)	-0.55	6.49	11.89	(-3.54, +2.44)	(-24.01, +22.90)
V_{OC} (V)	+0.0244	0.0916	0.2307	(-0.0334, +0.0821)	(-0.4288, +0.4775)
J_{SC} (mA cm⁻²)	+1.3222	1.9173	2.3526	(+0.8323, +1.8121)	(-2.5218, +5.1663)
PCE_{MPPT} (%)	+0.4628	1.9145	2.6241	(-0.5572, +1.4827)	(-4.6928, +5.6184)

Table 3: Agreement statistics for SMS vs. commercial system (commercial system holder on both systems).

Metric	Mean Difference	MAE	RMSE	95% CI	95% Limits of Agreement
Fill factor (%)	-2.21	5.29	6.94	(-3.89, -0.52)	(-15.22, +10.81)
V_{OC} (V)	+0.0049	0.0158	0.0283	(-0.0022, +0.0121)	(-0.0502, +0.0600)
J_{SC} (mA cm⁻²)	-0.3306	0.4229	0.7264	(-0.4962, -0.1650)	(-1.6086, +0.9474)
PCE_{MPPT} (%)	-0.4478	0.9822	1.6322	(-1.0439, +0.1483)	(-3.5767, +2.6811)

Table 4: Platform repeatability statistics for fill factor and power.

Metric	Fill factor (%)	Power (mW, MPPT derived)
Repeatability σ_r	0.0087	0.0331
RSD (%)	1.26%	0.34%
95% Repeatability Coefficient	0.0241	0.0918

$\pm 13\%$, and the RMSE for J_{SC} was reduced by a factor of three. This demonstrates that the custom 3D-printed sample holder, particularly the variability in contact pressure from its spring-loaded pogo pins, was a dominant source of random error. The commercial holder provides more consistent contact and alignment, thereby reducing the measurement scatter. This experiment effectively decouples the performance of the measurement electronics from the sample interface, confirming that the core electronic components of the SMS are reliable and that the primary source of measurement variability lies in the mechanical design of the sample holder.

Repeatability Analysis

To evaluate the intrinsic precision of the SMS, we performed four consecutive JV and MPPT measurements on a single PSC substrate. As summarized in Table 4, the relative standard deviations (RSD) were 1.26% for fill factor (FF) and 0.34% for maximum power conversion efficiency (PCE_{MPPT}). These low RSD values confirm that the SMS provides highly repeatable results when external variables like substrate handling are minimized.

Establishing this high level of hardware precision is a critical first step, as it allows for the differentiation of equipment-related error from method-induced variability. In the first major inter-laboratory comparison of perovskite solar devices, Dunbar et al. found that for a slowly responding device, the efficiency measurements between labs varied by as much as 37% [29]. They concluded that this enormous discrepancy was primarily caused by the choice of measurement method (including pre-conditioning, scan speed, and scan direction), which was "vastly more important than temperature control or irradiance calibration". Our repeatability measurement, with a PCE_{MPPT} variability of only 0.34%, demonstrates that our hardware is not a significant source of this error and serves as a reliable platform for investigating the procedural challenges that the study by Dunbar et al. identified as the central cause of measurement variability in the field.

The 200-hour degradation test under LED illumination demonstrated the SMS's suitability for reliable long-term monitoring. The system successfully maintained continuous operation and data logging. The use of silver paste contacts facilitated stable measurements over the duration of the test. The observed degradation trends were clear, which validates the system's utility for stability assessments under controlled laboratory conditions. To validate this result, we compared the SMS result with a different commercial system, which was designed for long term stability testing in [4]. The amount of noise is similar, as

is the variability in PCE_{MPPT} across the time of the study.

5 Future Work

Several enhancements could further improve the SMS's measurement accuracy and expand its capabilities for comprehensive stability testing, including: environmental control, improving the sample holder, and using higher precision electronics.

Integrating the SMS with open-source temperature and humidity chambers [8] would allow for the systematic investigation of thermal and moisture stress effects on perovskite solar devices. This would create a complete environmental testing platform, while maintaining a cost-effective approach, and would enable researchers to study degradation mechanisms under realistic operating conditions on a massive scale. Along these lines, a more robust sample holder with an improved mechanical design could reduce the contact pressure variability observed in our experiments. Potential improvements include: a uniform spring-loaded contact mechanism with calibrated force application, better alignment features to ensure reproducible device positioning, and a more reliable N_2 purging system. These modifications would significantly reduce the measurement variability observed when transferring samples between different holders.

A key limitation of the current SMS is its two-wire sensing configuration, where the same wires are used to supply current and measure voltage. This approach can introduce measurement variability from contact resistance, potentially leading to inaccurate current readings. While the two wire design was initially chosen to reduce hardware and firmware complexity, implementing a four-wire Kelvin sensing architecture [30] would be a crucial next step for creating a more accurate version of the system. By separating the current-carrying and voltage-sensing paths, this technique would provide more precise voltage measurements, particularly for low-resistance substrates. Upgrading to more accurate current and voltage sensing integrated circuits would improve measurement resolution and reduce systematic errors. Higher-resolution ADCs (16-bit or greater) and precision current sensors with lower offset drift would enhance the system's ability to detect subtle changes in substrate performance during long-term stability studies. This would be particularly beneficial for tracking gradual degradation trends that might be masked by the current levels of measurement noise.

6 Conclusion

We have developed and validated an open-source, Arduino-based Stability Measurement System (SMS) for high-throughput solar device stability analysis. At approximately 1% of the cost of commercial equipment, the system significantly lowers the financial barrier for comprehensive stability testing. The modular system performs parallel current-voltage characterization and maximum power point tracking on numerous devices simultaneously, enabling large-scale screening experiments. Validation against a commercial analyzer confirmed the SMS provides sufficient accuracy for comparative stability studies, with relative standard deviations of 1.26% for fill factor and 0.34% for power conversion efficiency. While not intended to replace high-precision commercial systems, the SMS is an effective tool for preliminary screening of materials and device architectures as well as longer term stability studies. By making high-throughput aging studies more accessible, this open-source platform can facilitate the accelerated development and eventual commercialization of stable perovskite solar devices.

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